

Molecular Dynamics: Interactions and Force Fields

PWF Summer school on Theoretical and Computational Biophysics

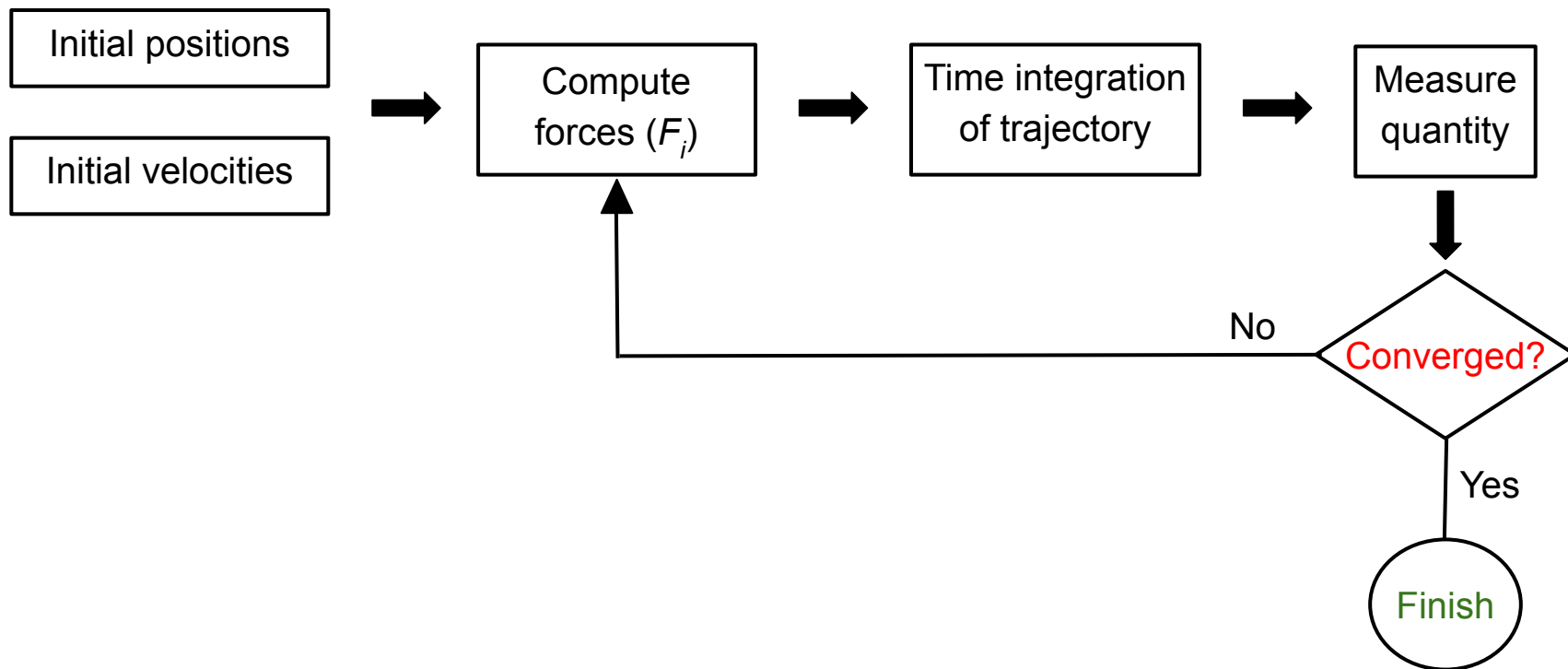
University of Colombo

Day 2/5

Dr. Marta Monti

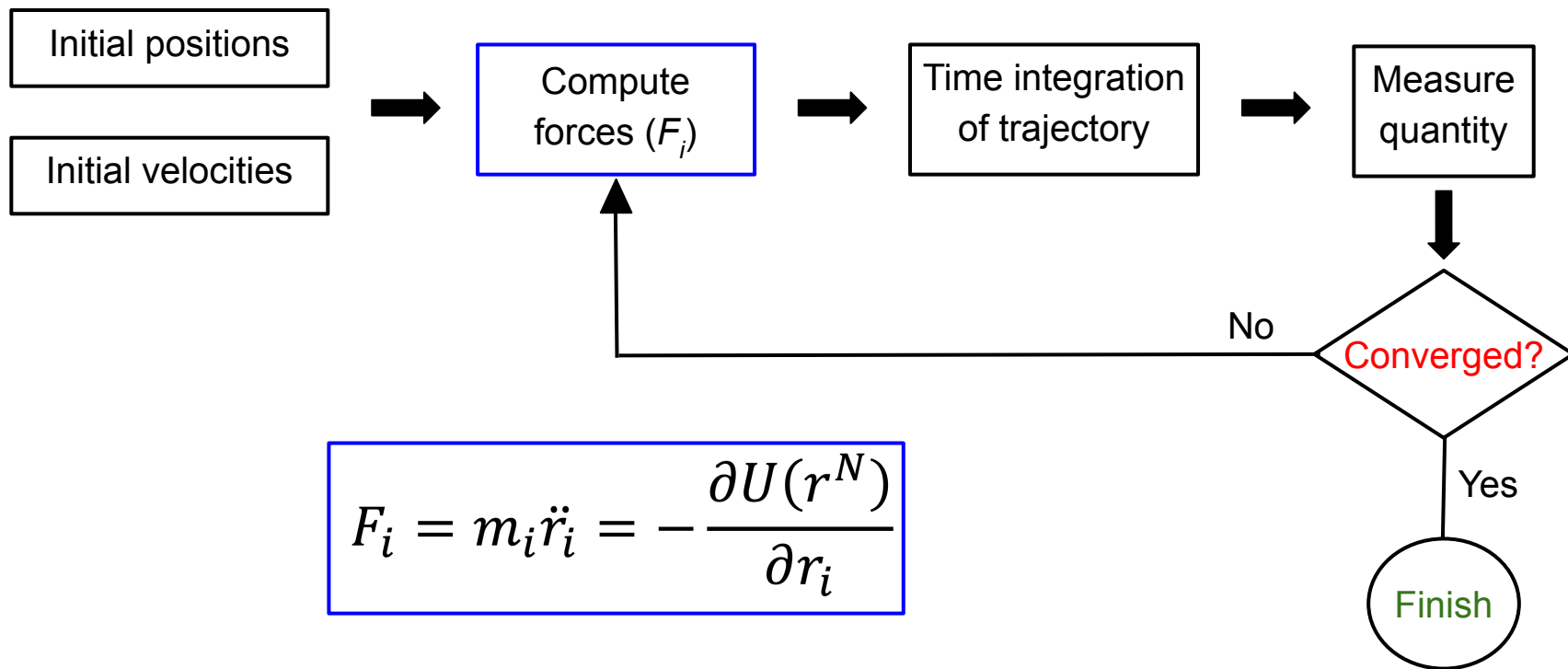
Overview

In a molecular dynamics (MD) simulation, we integrate the equation of motions to follow the temporal evolution of a set of N **interacting** particles:



Overview

In a molecular dynamics (MD) simulation, we integrate the equation of motions to follow the temporal evolution of a set of N **interacting** particles:



Interatomic interactions

$$F_i = m_i \ddot{r}_i = - \frac{\partial U(r^N)}{\partial r_i}$$

Force acting on the i particle due to its interaction with other particles

We derive the forces from the potential energy

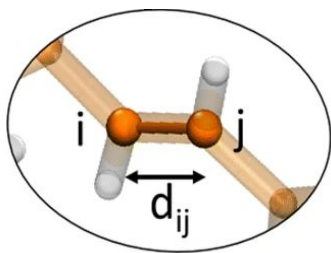
The potential energy function can be split into three components: **bonded** term, **non-bonded** term, and the term due to the presence of an **external field**:

$$U(r^N) = U_{bond.} + U_{non-bond.} + U_{ext.field}$$

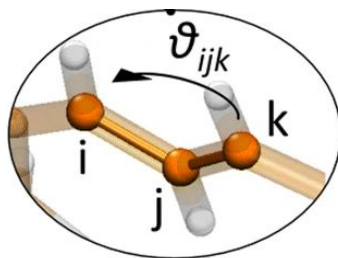
Bonded term

The bonded term includes the 2-, 3-, and 4-body interactions between atoms that are **bonded covalently**.

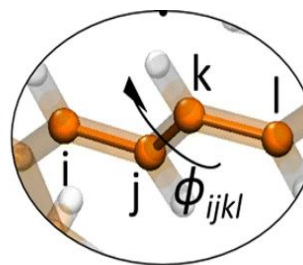
bonds



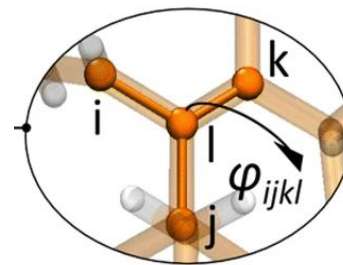
angles



dihedrals



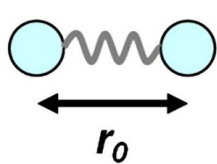
proper
dihedral



improper
dihedral

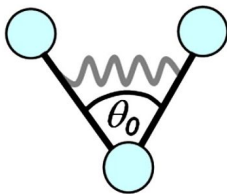
Bonded term

The bond stretching between two atoms, i and j , is usually described with a harmonic potential



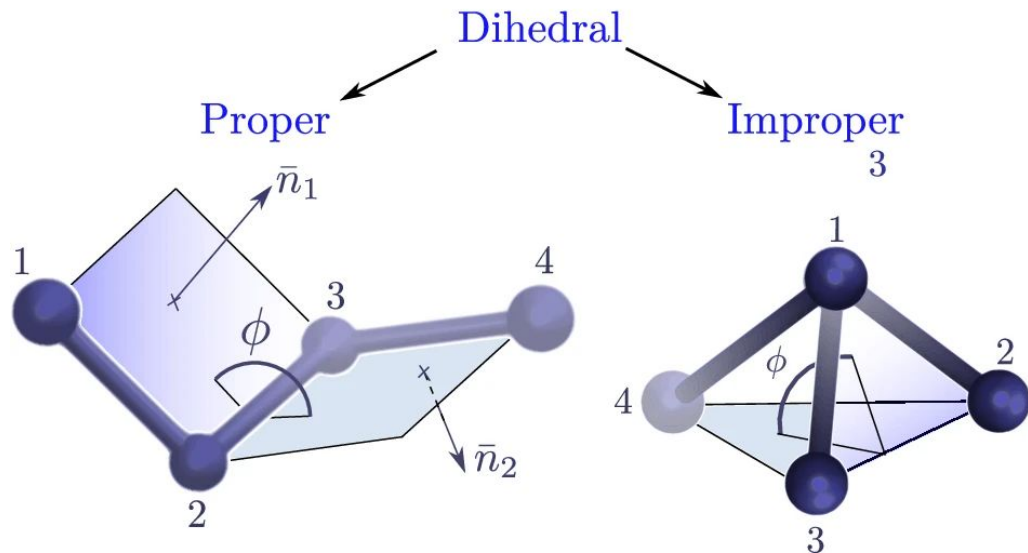
$$U_{bo}(r_{ij}) = \frac{1}{2}k_{bo,ij} \left(r_{ij} - r_{ij}^0 \right)^2$$

The angle bending between a triplet of i - j - k atoms is also described with a harmonic potential



$$U_a(\Theta_{ijk}) = \frac{1}{2}k_{a,ijk} \left(\Theta_{ijk} - \Theta_{ijk}^0 \right)^2$$

Bonded term



Proper dihedrals can be described by a sum of cosine functions:

$$U_d(\phi_{ijkl}) = \sum_n k_{d,ijkl} [1 + \cos(n\phi_{ijkl} - \delta)]$$

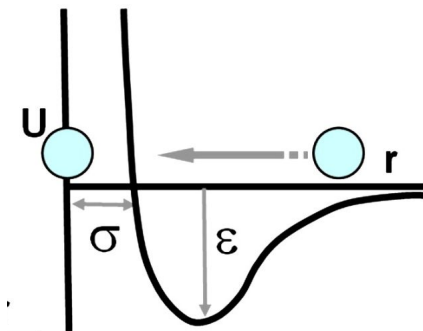
Improper dihedrals are described by means of harmonic potential functions:

$$U_{id}(\zeta_{ijkl}) = \frac{1}{2} k_{id,ijkl} (\zeta_{ijkl} - \zeta_{ijkl}^0)^2$$

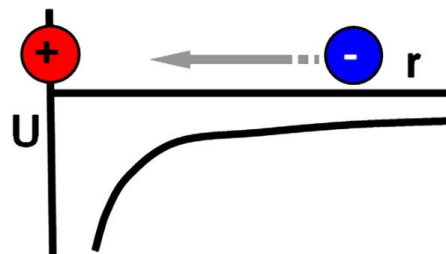
Non-bonded term

- The non-bonded term should consider all the potential terms between non-bonded atoms.
- We approximate it considering only the interaction between pair of particles → **pairwise potential**

van der Waals
term

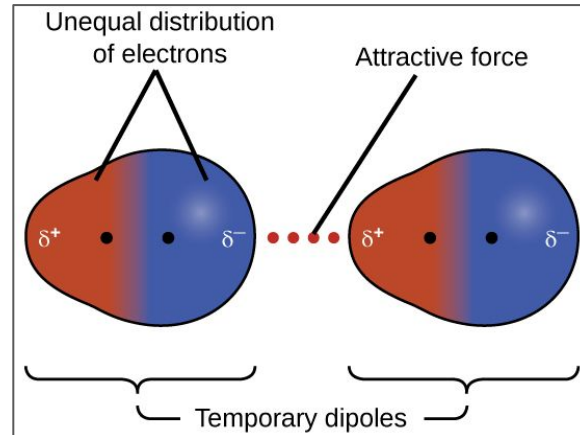


electrostatic
term



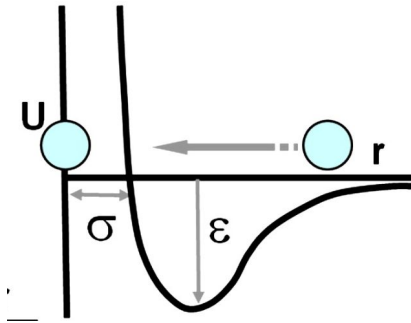
van der Waals

- In classical MD, the **van der Waals term** groups the non-bonded, non-electrostatic interactions between atoms: **London dispersion attraction** and **short-range Pauli repulsion**.
- London dispersion: temporary electron-density fluctuations create instantaneous dipoles that induce dipoles in nearby atoms, leading to attraction.
- Short-range repulsion: when atoms get too close, their electron clouds overlap, leading to strong Pauli repulsion.
- Both contributions depend strongly on the distance between atoms.



van der Waals

The van der Waals terms of the i, j atomic pairs can be modeled by the Lennard-Jones potential:



$$U_{LJ}(r) = 4\epsilon_{ij} \left[\underbrace{\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12}}_{\text{Repulsion}} - \underbrace{\left(\frac{\sigma_{ij}}{r_{ij}} \right)^6}_{\text{Attraction}} \right]$$

Repulsion:

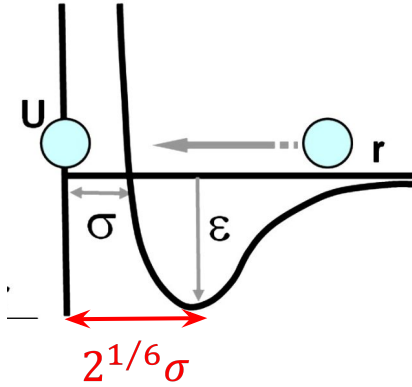
Pauli repulsion due to
electron-cloud overlap

Attraction:

London dispersion forces

van der Waals

The van der Waals terms of the i, j atomic pairs can be modeled by the Lennard-Jones potential:

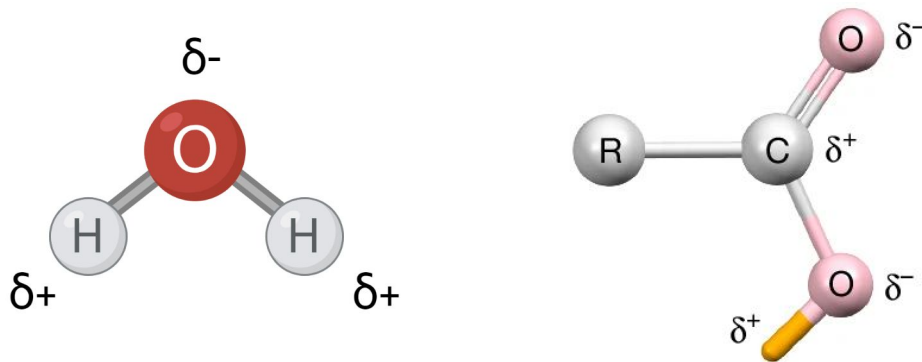


$$U_{LJ}(r) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]$$

- ϵ_{ij} = depth of the potential well $\rightarrow$ it controls the strength of the interaction
- σ_{ij} = distance at which the LJ potential is zero (if $r = \sigma \rightarrow U_{LJ}=0$)
- $r = 2^{1/6}\sigma \rightarrow U_{\min} = -\epsilon$

Electrostatics

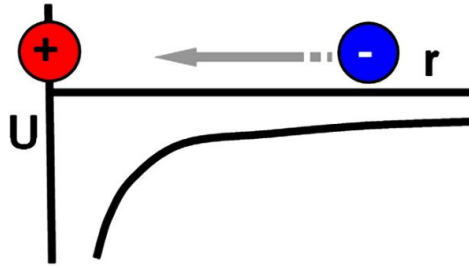
Atoms in molecules can carry partial positive or negative charges due to an uneven distribution of electrons.



- Electrostatic interactions arise between electrically charged or partially charged atoms
- Their strength depends on the **magnitude of the charges** and the **distance between them**
- In classical MD, atoms are treated as point charges carrying partial charges

Electrostatics

The electrostatic interactions can be modeled by the Coulomb potential:

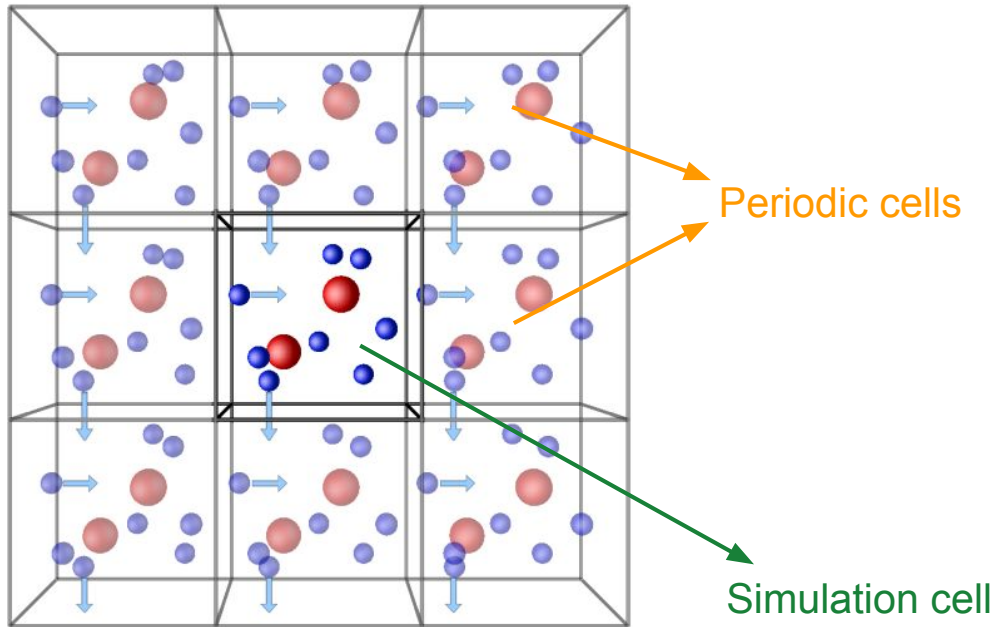


$$U_C = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}$$

- $q_i, q_j \rightarrow$ atomic partial charges
- $r_{ij} \rightarrow$ distance between atoms
- $q_i q_j < 0 \rightarrow$ attraction ($U < 0$); $q_i q_j > 0 \rightarrow$ repulsion ($U > 0$)
- magnitude decreases as $1/r \rightarrow$ unlike van der Waals interactions, electrostatics are **long-ranged**

What happens under periodic boundary conditions?

- The simulation box is replicated infinitely in all directions to mimic a bulk system
- Each atom interacts not only with atoms in the original box, but also with their periodic images



In a cubic simulation cell of volume L^3 :

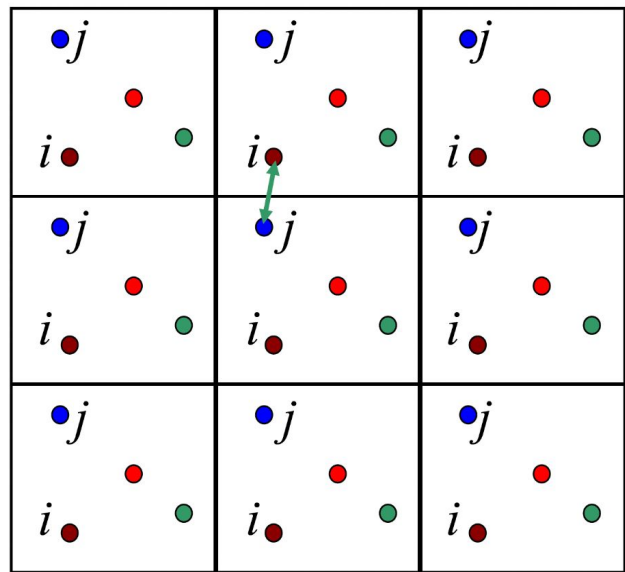
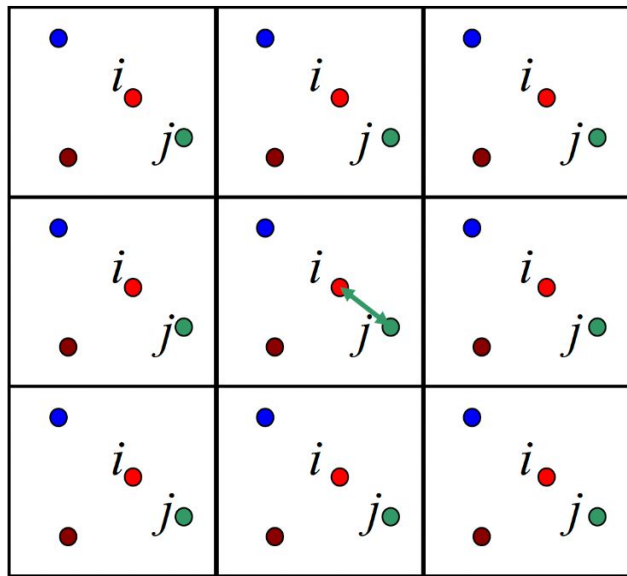
$$\left\{ \begin{array}{l} \text{position of particle } i \\ \text{in replica indexed} \\ \text{by } \mathbf{n} = (n_x, n_y, n_z) \end{array} \right\} = \mathbf{r}_{i,\mathbf{n}} = \mathbf{r}_i + \mathbf{n}L$$

position of i in
primary cell

$\mathbf{n} = (0,0,0)$
primary cell

Minimum image convention

When potential decays rapidly with distance (= short-range interactions):
we neglect all $\mathbf{n}$ except the one for which $|\mathbf{r}_i - \mathbf{r}_j + \mathbf{n}L|$ is the smallest



Minimum image convention

When potential decays rapidly with distance...



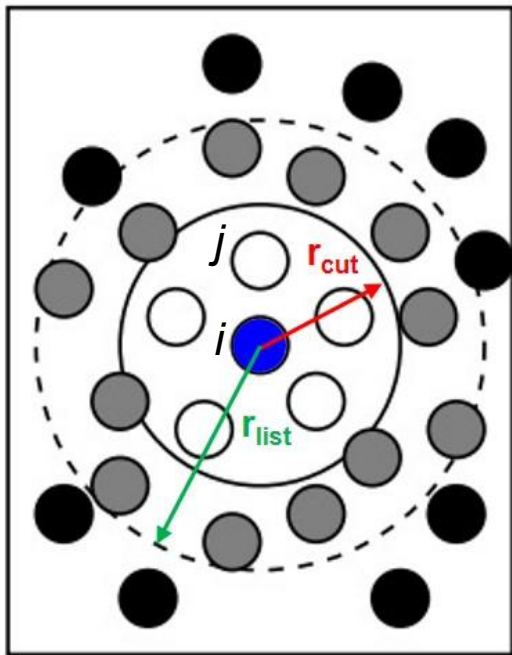
Lennard-Jones potential

$$U_{LJ}(r) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]$$

- Short-range forces using the minimum image convention and PBC require the calculation of $N(N-1)/2$ distances and pair interactions for each type of potential → **huge number**
- Usually short-range potentials can be safely cut off at a distance shorter than $L/2$.

Verlet neighbor lists

Calculation of $N(N-1)/2$ distances can be avoided at most simulation steps using neighbor list



- **Verlet scheme:** build a neighbor list containing all atoms within r_{list}
- If $r_{ij} < r_{\text{cut}} \rightarrow$ compute the interaction
- The neighbor list is updated periodically as atoms move
- $r_{\text{list}} > r_{\text{cut}}$ provides a buffer so no interaction is missed between updates

Long-range interactions

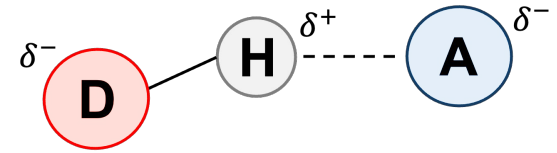
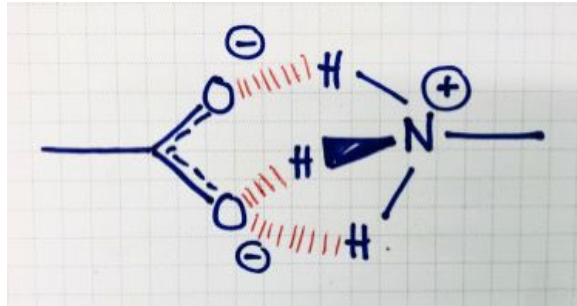
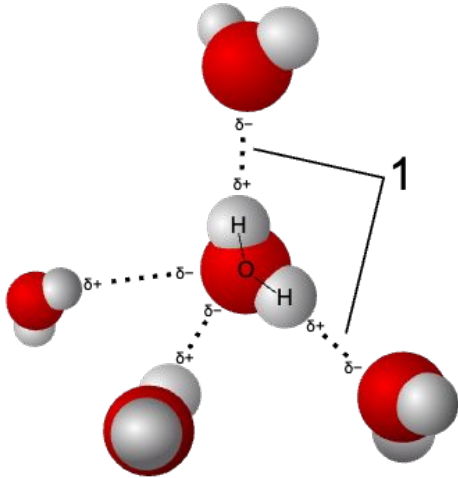
These approximations don't work for long-range interactions (Coulomb potential)

$$U_{C,tot} = \frac{1}{4\pi\epsilon_0} \frac{1}{2} \sum_{\mathbf{n}} \sum_{ij} \frac{q_i q_j}{r_{ij,\mathbf{n}}}$$

- Interactions between the particle and its images in the periodic cells should be explicitly included in the potential calculation
- **Full sum** is slowly (and conditionally) convergent → **too computationally expensive**
- First solution: Ewald method (scales as N^2); current solution: **Particle-mesh Ewald** (PME) method (scales as $N\log(N)$)

Hydrogen bonds

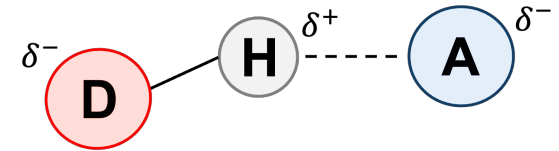
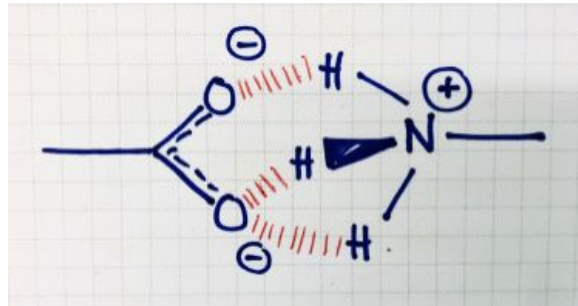
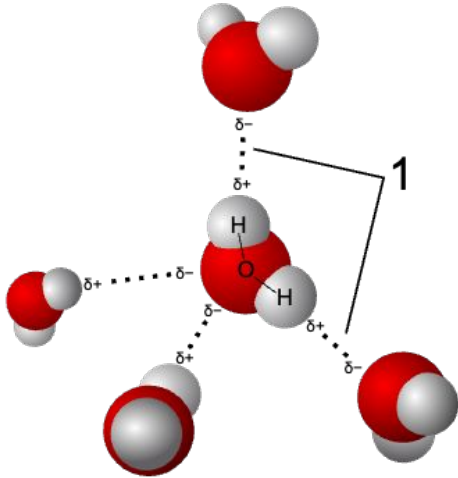
- Molecular interaction where an H atom bonded to a highly electronegative atom (= **donor**) is attracted to another electronegative atom (= **acceptor**) bearing a lone pair of electrons
- H-bonds play a fundamental role in biology and chemistry: they are responsible for the anomalously high boiling point of water and the stabilization of proteins



$$d(D \cdots A) \lesssim 3.0 \text{ \AA}$$
$$a(D-H \cdots A) \gtrsim 150^\circ$$

Hydrogen bonds

- They are not represented by an explicit H-bond potential
- H-bonds emerge from the combination of: a) electrostatic interactions between partial charges and b) van der Waals interactions between donor and acceptor atoms
- Their strength depends on the partial charges and LJ parameters



$$d(D \cdots A) \lesssim 3.0 \text{ \AA}$$
$$a(D-H \cdots A) \gtrsim 150^\circ$$

Force Field

All these potential terms are collected together in the **force field** (FF) that we provide in our MD simulation.

Practical example of an FF in GROMACS (OPLS-AA):

ffbonded.itp

(bonds, angles, and dihedrals)

```
[ bondtypes ]
; i      j      func      b0          kb          ; For TIP4F Water - wlj 1/98
OW      HW      1         0.09572    502080.0    ;
OW      LP      1         0.01750    753120.0    ; -idem-
C*      HC      1         0.10800    284512.0    ;
C       C3      1         0.15220    265265.6    ; END
C_2     C3      1         0.15220    265265.6    ; END
C_3     C3      1         0.15220    265265.6    ; END
C       CA      1         0.14900    334720.0    ; wlj 8/97
C_2     CA      1         0.14900    334720.0    ; wlj 8/97
C_3     CA      1         0.14900    334720.0    ; wlj 8/97
C       CB      1         0.14190    374049.6    ; GUA
```

ffnonbonded.itp

(partial charges, LJ parameters)

```
[ atomtypes ]
; full atom descriptions are available in ffoplsaa.atp
; name      bond_type      mass      charge      ptype      sigma      epsilon
opls_001    C      6      12.01100    0.500      A      3.75000e-01  4.39320e-01 ;
opls_002    O      8      15.99940   -0.500      A      2.96000e-01  8.78640e-01 ;
opls_003    N      7      14.00670   -0.570      A      3.25000e-01  7.11280e-01 ;
opls_004    H      1      1.00800    0.370      A      0.00000e+00  0.00000e+00 ;
opls_005    C2     6      14.02700    0.200      A      3.80000e-01  4.93712e-01 ;
opls_006    CH     6      13.01900    0.200      A      3.80000e-01  3.34720e-01 ;
opls_007    C3     6      15.03500    0.000      A      3.91000e-01  6.69440e-01 ;
opls_008    CH     6      13.01900    0.000      A      3.85000e-01  3.34720e-01 ;
opls_009    C2     7      14.02700    0.000      A      3.90500e-01  4.93712e-01 ;
opls_010    C3     6      15.03500    0.000      A      3.90500e-01  7.32200e-01 ;
```

b0 = equilibrium bond distance

kb = spring constant → stiffness of the bond

Force Field

- Parameters of the FFs are usually obtained by combining quantum mechanical calculations, experimental data, and fitting procedures.
- The quality of an FF is usually assessed by its ability to reproduce thermodynamic properties of bulk solutions
- Typical FFs for biomolecules are: AMBER, CHARMM, and OPLS-AA

We can distinguish all-atom and united-atom FFs



Interatomic potential parameters are defined for each atom type



Atoms of the same chemical group (e.g. H and C atoms of CH₃) are treated as one interaction center.

Force Field

What about the parameters for H_2O ?

